

Instituto Tecnológico de Costa Rica

Campus Tecnológico Local San José

Laboratorio 06: Cursores

Profesor:

Adriana Álvarez Figueroa

Estudiantes:

González Prendas Mauricio: 2024143009

Hidalgo Paz Carmen: 2020030538

Meza Marin Dylan: 2021577352

Fecha de entrega:

29/04/25

Semestre I 2025

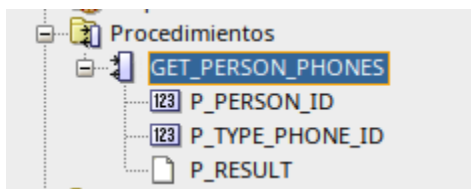
1. Cree un procedimiento (utilizando OPEN del cursor) que retorne todos los teléfonos de una persona. El número de persona debe ser enviado como parámetro así como el tipo de teléfono y este último puede ser nulo, en cuyo caso se deben desplegar todos los teléfonos de la persona. 15pts

Este procedimiento, dado el ID de una persona, abre un SYS_REFCURSOR con sus teléfonos; permite un filtro opcional por tipo, y devuelve para cada registro el ID, el número y el nombre del tipo de teléfono

Codigo:

```
CREATE OR REPLACE PROCEDURE ge.get_person_phones (  
    p_person_id      IN ge.people.id%TYPE,  
    p_type_phone_id  IN ge.type_phone.id%TYPE DEFAULT NULL,  
    p_result         OUT SYS_REFCURSOR  
) IS  
BEGIN  
    /* Abre un cursor explícito con los teléfonos que  
       pertenecen a la persona. Si p_type_phone_id es  
       nulo, se devuelven todos los teléfonos */  
    OPEN p_result FOR  
        SELECT ph.id,  
               ph.phone_number,  
               tp.name AS type name  
        FROM   ge.phonexpeople pxp  
        JOIN   ge.phone         ph ON ph.id = pxp.id_phone  
        JOIN   ge.type_phone    tp ON tp.id = ph.id_type_phone  
        WHERE  pxp.id_people = p_person_id  
               AND (p_type_phone_id IS NULL  
                   OR ph.id_type_phone = p_type_phone_id);  
END get_person_phones;  
/
```

Resultado con el procedimiento creado:



2. Cree un test para probar el procedimiento utilizando cursores. 20pts

Este script de prueba en PL/SQL invoca ge.get_person_phones con un cursor de salida: primero muestra solo los teléfonos tipo "Celular" (ID 2) de la persona 2 y luego lista todos sus números, formateando cada fila (ID, número y tipo) con DBMS_OUTPUT para validación rápida en consola.

Codigo:

```

SET SERVEROUTPUT ON;
DECLARE
    v_rc          SYS_REFCURSOR;
    v_id          ge.phone.id%TYPE;
    v_number      ge.phone.phone_number%TYPE;
    v_type_name   ge.type_phone.name%TYPE;
BEGIN
    DBMS_OUTPUT.put_line('--- Prueba #1: Teléfonos Celular de la persona 2 ---');

    -- El tipo "Celular" tiene id = 2
    ge.get_person_phones(
        p_person_id    => 2,
        p_type_phone_id => 2,
        p_result       => v_rc
    );

    LOOP
        FETCH v_rc INTO v_id, v_number, v_type_name;
        EXIT WHEN v_rc%NOTFOUND;
        DBMS_OUTPUT.put_line(
            RPAD(v_id,5)||' '||RPAD(v_number,15)||' '||v_type_name
        );
    END LOOP;
    CLOSE v_rc;

    DBMS_OUTPUT.put_line(CHR(10)||'--- Prueba #2: TODOS los teléfonos de la persona 2 ---');

    -- Esta vez sin filtrar por tipo
    ge.get_person_phones(
        p_person_id    => 2,
        p_type_phone_id => NULL,
        p_result       => v_rc
    );

    LOOP
        FETCH v_rc INTO v_id, v_number, v_type_name;
        EXIT WHEN v_rc%NOTFOUND;
        DBMS_OUTPUT.put_line(
            RPAD(v_id,5)||' '||RPAD(v_number,15)||' '||v_type_name
        );
    END LOOP;
    CLOSE v_rc;
END;

```

Resultado:

```

--- Prueba #1: Teléfonos Celular de la persona 2 ---
2      555-5678      Celular

--- Prueba #2: TODOS los teléfonos de la persona 2 ---
2      555-5678      Celular
6      555-2345      Oficina

Procedimiento PL/SQL terminado correctamente.

```

3. ¿Qué pasa cuando en el fetch del test se agregan más campos? 10pts

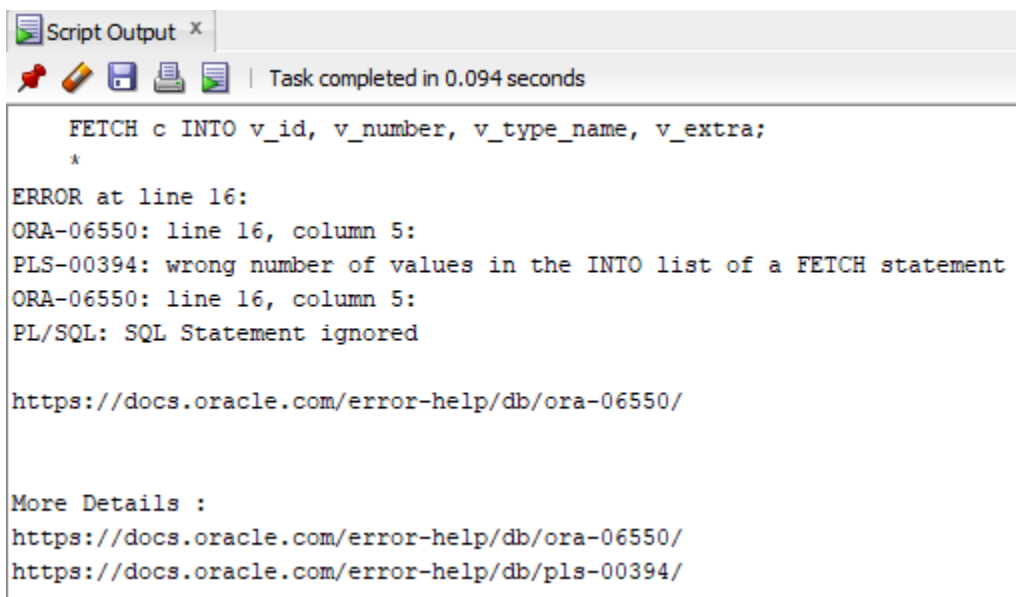
Cuando se intenta agregar más campos en el fetch ocurre un error. Este es el código con del fetch:

```
SET SERVEROUTPUT ON;
DECLARE
    CURSOR c IS
        SELECT p.id, p.phone_number, tp.name
        FROM ge.phone p
        JOIN ge.type_phone tp ON p.id_type_phone = tp.id
        WHERE p.id = 2 AND p.id_type_phone = 2;

    v_id      ge.phone.id%TYPE;
    v_number  ge.phone.phone_number%TYPE;
    v_type_name ge.type_phone.name%TYPE;
    v_extra   VARCHAR2(100); -- variable adicional
BEGIN
    DBMS_OUTPUT.put_line('--- Prueba Punto 3: MÁS variables que columnas ---');

    OPEN c;
    FETCH c INTO v_id, v_number, v_type_name, v_extra;
    CLOSE c;
END;
/
```

Este es el error mostrado:



The screenshot shows a 'Script Output' window with a toolbar at the top. The status bar indicates 'Task completed in 0.094 seconds'. The main text area displays the following error message:

```
FETCH c INTO v_id, v_number, v_type_name, v_extra;
*
ERROR at line 16:
ORA-06550: line 16, column 5:
PLS-00394: wrong number of values in the INTO list of a FETCH statement
ORA-06550: line 16, column 5:
PL/SQL: SQL Statement ignored

https://docs.oracle.com/error-help/db/ora-06550/

More Details :
https://docs.oracle.com/error-help/db/ora-06550/
https://docs.oracle.com/error-help/db/pls-00394/
```

4. ¿Qué pasa cuando en el fetch del test faltó obtener campos? 10pts

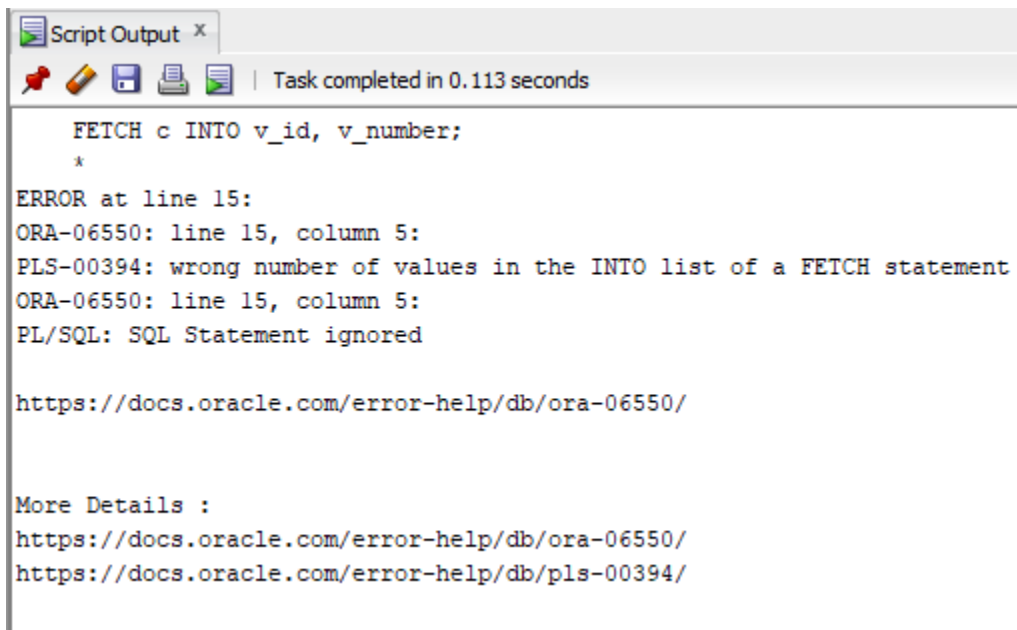
Cuando se omite en el fetch un campo se da un error. Este es el código del fetch:

```
SET SERVEROUTPUT ON;
DECLARE
  CURSOR c IS
    SELECT p.id, p.phone_number, tp.name
    FROM ge.phone p
    JOIN ge.type_phone tp ON p.id_type_phone = tp.id
    WHERE p.id = 2 AND p.id_type_phone = 2;

  v_id      ge.phone.id%TYPE;
  v_number  ge.phone.phone_number%TYPE;
  -- v_type_name ge.type_phone.name%TYPE; -- NO se declara esta variable para simular el error
BEGIN
  DBMS_OUTPUT.put_line('--- Prueba Punto 4: MENOS variables que columnas ---');

  OPEN c;
  FETCH c INTO v_id, v_number;
  CLOSE c;
END;
/
```

Este es el error mostrado:



The screenshot shows a 'Script Output' window with a toolbar at the top. Below the toolbar, it says 'Task completed in 0.113 seconds'. The main text area displays the following error message:

```
FETCH c INTO v_id, v_number;
*
ERROR at line 15:
ORA-06550: line 15, column 5:
PLS-00394: wrong number of values in the INTO list of a FETCH statement
ORA-06550: line 15, column 5:
PL/SQL: SQL Statement ignored

https://docs.oracle.com/error-help/db/ora-06550/

More Details :
https://docs.oracle.com/error-help/db/ora-06550/
https://docs.oracle.com/error-help/db/pls-00394/
```

5. ¿Qué pasa cuando en el fetch del test los campos tienen diferente dominio del que está obteniendo el cursor? 10pts

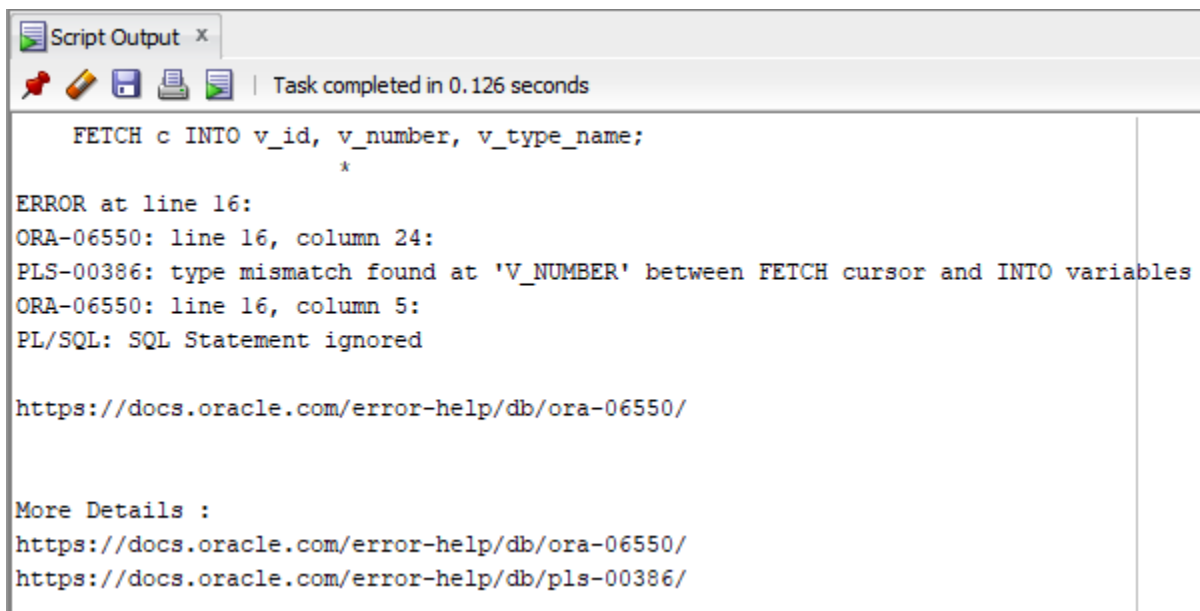
Cuando los campos en el fetch tienen tipos incorrectos, se genera un error. Este es el código del fetch:

```
SET SERVEROUTPUT ON;
DECLARE
    CURSOR c IS
        SELECT p.id, p.phone_number, tp.name
        FROM ge.phone p
        JOIN ge.type_phone tp ON p.id_type_phone = tp.id
        WHERE p.id = 2 AND p.id_type_phone = 2;

    -- X Tipos incorrectos a propósito
    v_id      VARCHAR2(100); -- debería ser NUMBER
    v_number   BOOLEAN;      -- debería ser VARCHAR2
    v_type_name BOOLEAN;      -- debería ser VARCHAR2
BEGIN
    DBMS_OUTPUT.put_line('--- Prueba Punto 5: TIPOS DE DATOS INCOMPATIBLES ---');

    OPEN c;
    FETCH c INTO v_id, v_number, v_type_name;
    CLOSE c;
END;
/
```

Aquí está el error generado:



The screenshot shows a 'Script Output' window with a toolbar at the top containing icons for pin, edit, save, print, and a status bar indicating 'Task completed in 0.126 seconds'. The main text area displays the following error message:

```
FETCH c INTO v_id, v_number, v_type_name;
          *
ERROR at line 16:
ORA-06550: line 16, column 24:
PLS-00386: type mismatch found at 'V_NUMBER' between FETCH cursor and INTO variables
ORA-06550: line 16, column 5:
PL/SQL: SQL Statement ignored

https://docs.oracle.com/error-help/db/ora-06550/

More Details :
https://docs.oracle.com/error-help/db/ora-06550/
https://docs.oracle.com/error-help/db/pls-00386/
```

6. Modifique el cursor para utilizar el código con LOOP. 10pts

```

--=====
-- Punto 6
--=====

CREATE OR REPLACE PROCEDURE ge.get_person_phones_loop (
    p_person_id      IN  ge.people.id%TYPE,
    p_type_phone_id  IN  ge.type_phone.id%TYPE DEFAULT NULL
) IS
    CURSOR cur_phones IS
        SELECT ph.id,
               ph.phone_number,
               tp.name AS type_name
        FROM   ge.phonexpeople pxp
        JOIN   ge.phone      ph ON ph.id = pxp.id_phone
        JOIN   ge.type_phone tp ON tp.id = ph.id_type_phone
        WHERE  pxp.id_people = p_person_id
              AND (p_type_phone_id IS NULL
                  OR ph.id_type_phone = p_type_phone_id);

    v_id      ge.phone.id%TYPE;
    v_number  ge.phone.phone_number%TYPE;
    v_type_name ge.type_phone.name%TYPE;
BEGIN
    OPEN cur_phones;
    LOOP
        FETCH cur_phones INTO v_id, v_number, v_type_name;
        EXIT WHEN cur_phones%NOTFOUND;
        DBMS_OUTPUT.PUT_LINE('ID: ' || v_id || ', Tel: ' || v_number || ', Tipo: ' || v_type_name);
    END LOOP;
    CLOSE cur_phones;
END;

```

Script Output x

Task completed in 0.209 seconds

Procedure GE.GET_PERSON_PHONES_LOOP compiled

7. Modifique el cursor para utilizar el código de SYS_REFCURSOR.10pts

```

File Edit View Navigate Run Source Team Tools Window Help
Connections Oracle Connections
GE
Tables (Filtered)
ID
FIRST_NAME
SECOND_NAME
FIRST_SURNAME
SECOND_SURNAME
SALARY
BIRTHDAY
ID_TYPE_PERSON
ID
PHONE_NUMBER
ID_TYPE_PHONE
PHONEPEOPLE
PHONEPEOPLE
PRODUCT
PRODUCTCART
TYPE_PERSON
TYPE_PHONE

Reports
All Reports
About Your Database
All Objects
Application Express
ASH and AWR
Database Administration
Data Dictionary
Data Dictionary Reports

Welcome Page 07.1ad 0.065 seconds

Worksheet Query Builder
1 CREATE OR REPLACE PROCEDURE ge.get_People_Phone (p_id_person IN ge.people.id%TYPE, p_kindofphone_name IN ge.type_phone.name%TYPE DEFAULT NULL) AS
2
3     v_cursor SYS_REFCURSOR;
4     v_firstname ge.people.first_name%TYPE;
5     v_number ge.phone.phone_number%TYPE;
6     v_type_name ge.type_phone.name%TYPE;
7 BEGIN
8     -- Abrir cursor con telefonos de la persona, filtrando por tipo si se indica
9     OPEN v_cursor FOR
10        SELECT p.first_name, ph.phone_number, kp.name AS kind
11        FROM ge.phonexpeople pxp
12        INNER JOIN ge.phone ph ON ph.id = pxp.id_phone
13        INNER JOIN ge.people p ON p.id = pxp.id_people
14        INNER JOIN ge.type_phone kp ON kp.id = ph.id_type_phone
15        WHERE p.id = p_id_person
16              AND (kp.name = NVL(p_kindofphone_name, kp.name));
17
18     -- Muestra resultados
19     LOOP
20         FETCH v_cursor INTO v_firstname, v_number, v_type_name;
21         EXIT WHEN v_cursor%NOTFOUND;
22         DBMS_OUTPUT.put_line('Nombre: ' || v_firstname ||
23                               ' | Telefono: ' || v_number ||
24                               ' | Tipo: ' || v_type_name);
25     END LOOP;
26     CLOSE v_cursor;
27 END;

```

Script Output x

Task completed in 0.065 seconds

Procedure GE.GET_PEOPLE_PHONE compiled

8.2. Creacion del proceso getCursos

The screenshot displays the Oracle SQL Developer interface. On the left, the 'Connections' pane shows the 'UN' database selected, with a tree view of its tables including COURSE, STATUS, and STUDENT. The main workspace is divided into a 'Worksheet' and a 'Query Builder'. The 'Worksheet' contains a PL/SQL procedure definition for UN.getCursos, which takes p_id_student and p_estado as input parameters and returns a cursor c_resultados. The procedure uses a SELECT statement with JOINs to retrieve student and course information based on the provided parameters. The 'Script Output' pane at the bottom shows the message 'Procedure UN.GETCURSOS compiled' and indicates that the task was completed in 0.174 seconds.

```
1 CREATE OR REPLACE PROCEDURE UN.getCursos (  
2   p_id_student    IN UN.STUDENT.id_student%TYPE,  
3   p_estado        IN UN.STATUS.status_name%TYPE,  
4   c_resultados    OUT SYS_REFCURSOR  
5 ) AS  
6 BEGIN  
7   OPEN c_resultados FOR  
8     SELECT s.id_student,  
9           c.course_name,  
10          st.status_name  
11   FROM UN.STUDENT s  
12   JOIN UN.STUDENT_X_COURSE sc ON s.id_student = sc.id_student  
13   JOIN UN.COURSE c ON sc.id_course = c.id_course  
14   JOIN UN.STATUS st ON st.id_studentxcourse = sc.id  
15   WHERE s.id_student = p_id_student  
16         AND (st.status_name = NVL(p_estado, st.status_name));  
17 END;  
18 /  
19
```

Script Output x
Task completed in 0.174 seconds
Procedure UN.GETCURSOS compiled

8.3. Prueba del proceso getCursos

The screenshot displays the Oracle SQL Developer interface. On the left, the 'Connections' pane shows the 'UN' connection selected, with a tree view of database objects including 'COURSE', 'STATUS', and 'STUDENT'. The 'Reports' pane is also visible. The main 'Worksheet' area contains a PL/SQL procedure named 'getCursos'. The procedure is designed to retrieve and display course information for a specific student (ID 1). It uses a cursor to fetch data from the 'COURSE' and 'STATUS' tables. The output is displayed in the 'Script Output' pane at the bottom.

```
1 SET SERVEROUTPUT ON;
2
3 DECLARE
4     v_cursor SYS_REFCURSOR;
5     v_id      UN.STUDENT.id_student%TYPE;
6     v_name    UN.COURSE.course_name%TYPE;
7     v_estado  UN.STATUS.status_name%TYPE;
8
9 BEGIN
10    -- Obtener cursos aprobados por el estudiante
11    DBMS_OUTPUT.put_line('--- Cursos Aprobados del Estudiante 1 ---');
12    UN.getCursos(1, 'Aprobado', v_cursor);
13    LOOP
14        FETCH v_cursor INTO v_id, v_name, v_estado;
15        EXIT WHEN v_cursor%NOTFOUND;
16        DBMS_OUTPUT.put_line('ID: ' || v_id || ' | Curso: ' || v_name || ' | Estado: ' || v_estado);
17    END LOOP;
18    CLOSE v_cursor;
19
20    -- Todos los cursos del estudiante
21    DBMS_OUTPUT.put_line(CHR(10) || '--- Todos los cursos del Estudiante 1 ---');
22    UN.getCursos(1, NULL, v_cursor);
23    LOOP
24        FETCH v_cursor INTO v_id, v_name, v_estado;
25        EXIT WHEN v_cursor%NOTFOUND;
26        DBMS_OUTPUT.put_line('ID: ' || v_id || ' | Curso: ' || v_name || ' | Estado: ' || v_estado);
27    END LOOP;
28    CLOSE v_cursor;
29 END;
```

Script Output

Task completed in 0.135 seconds

--- Cursos Aprobados del Estudiante 1 ---
ID: 1 | Curso: Matemáticas | Estado: Aprobado

--- Todos los cursos del Estudiante 1 ---
ID: 1 | Curso: Matemáticas | Estado: Aprobado
ID: 1 | Curso: Fisica | Estado: Pendiente

PL/SQL procedure successfully completed.